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AI-driven adaptive learning for sustainable educational transformation

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Abstract

This paper scrutinizes how adaptive learning technologies and artificial intelligence (AI) are transforming today's education by making it personalized, accessible, and efficient as well as leading people to accepting, addressing, and mitigating sustainable development. Recently, education witnessed a remarkable technological surge driven by various advances in technology, which can be demonstrated by the increase of the number of scientific publications on this topic from just 1 in 1990 to 636 in 2023. Ongoing digitalization and technological revolution in education together with the novel approach to respect each student's unique learning style and abilities paved the way for adaptive learning technologies represented by the innovative tools that personalize educational experiences to cater to individual learners. All of that contributes to preparing more educated and informed citizens, drives innovation, and supports economic growth necessary for achieving a sustainable future. Our bibliographic study employs VOSviewer to conduct a bibliometric analysis of a total number of 3518 selected publications using the keywords "adaptive learning" and "AI" (represented by articles, proceeding papers, and book chapters) indexed in the Web of Science (WoS) database from 1990 to 2024. Our results demonstrate that recent technological changes played a key role in transforming adaptive learning, which was rather reinforced by the "digital surge" in education brought about by the COVID-19 pandemic. Our findings can be useful for further development in the field of adaptive education where they can be employed by the relevant stakeholders and policy-makers as well as by the scholars and researchers.

KEYWORDS

adaptive learning, artificial intelligence, bibliographic analysis, sustainable development

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1 | INTRODUCTION

Education has always been an important aspect of the development of any society. It has also undergone numerous profound changes over time needed to reflect most recent challenges such as technological progress or global warming and climate change. (Akour & Alenezi, 2022; Alenezi, 2023). Nowadays, as we are entering the new post-COVID era that mark the transformation of the coronavirus pandemic (2020–2023) into the endemic, adaptive learning technologies, and artificial intelligence (AI) are revolutionizing education and its role in education for sustainability (EFS) like never before (Adiguzel et al., 2023; de Vries, 2022; Jing et al., 2023).

Adaptive learning technologies refer to educational systems that leverage data analytics and AI to personalize the learning experience. These technologies dynamically adjust the presentation of educational content based on an individual student's performance, learning pace, and preferences (Gligorea et al., 2023; Khan et al., 2022). By continuously analyzing data on student interactions, adaptive learning technologies provide customized support, targeted feedback, and optimized learning pathways to enhance student engagement and outcomes.

The arrival of AI in education marks a revolutionary shift, starting an era where personalized, adaptive learning becomes a reality. This transformative impact of AI paves the way for innovative approaches in sustainable education, reshaping traditional teaching methods into dynamic, learner-centered experiences. In the short run, the advent of AI might not change the education, but in the long run it would change it fundamentally.

Adaptive learning technologies and AI contribute to achieving sustainable future and it is quite straightforward to show how they do that: they help to tailor and provide education for all fostering a more education population capable of tackling complex global challenges, including sustainability. Better educated individuals make informed decisions about their daily lives and living environments, which in turn leads to more sustainable practices and policies. In addition, they promote innovation and solving most pressing global issues such as sustainability, climate change, and resource management. Finally, they help to shape up education that drives economic growth, which is crucial for funding and implementing sustainable initiatives.

The emergence of adaptive learning constitutes an evolutionary process that stems from centuries-long efforts to improve education. In ancient times, education was primarily limited to privileged elites who had access to personal tutors or attended prestigious institutions (Worth et al., 2023). Medieval universities used to be the places where the offspring of the elites would meet to study, mingle, make friends, and later in their lives strike powerful alliances for their further endeavors (Nureev et al., 2020). However, with time came advancements such as the printing press in the 15th century, which democratized knowledge dissemination through books and opened it to the wider masses (Zarifian et al., 2022). Then the Industrial Revolution in the 18th century brought about further changes in education as the need for basic literacy and numeracy skills became indispensable for an increasingly industrialized world (Purwanto et al., 2023). In

the 18th century, it was mass public schooling systems that opened education to the masses and introduced a major breakthrough. Most recently, the next breakthrough was driven by technology that once again helped to transform education. The advent of personal computers introduced interactive multimedia resources into classrooms while distance learning programs enabled students to access educational content remotely (Ferri et al., 2020). Finally, the spread of the Internet, first in the form of personal computing and then through mobile computing (i.e. using ubiquitous smartphones), made information and knowledge available for anyone virtually everywhere (with spatial computing being the next awaited breakthrough in education). Today, the main issue is not how to find some specific information but where to look for it and how to filter it and interpret the obtained data (Alashhab et al., 2021).

These developments laid the groundwork for adaptive learning technologies by demonstrating how technology could enhance educational experiences beyond traditional teaching methods. With it, AI and data-driven tools are applied to personalize and enhance (Lameras & Arnab, 2021; Tapalova & Zhiyenbayeva, 2022).

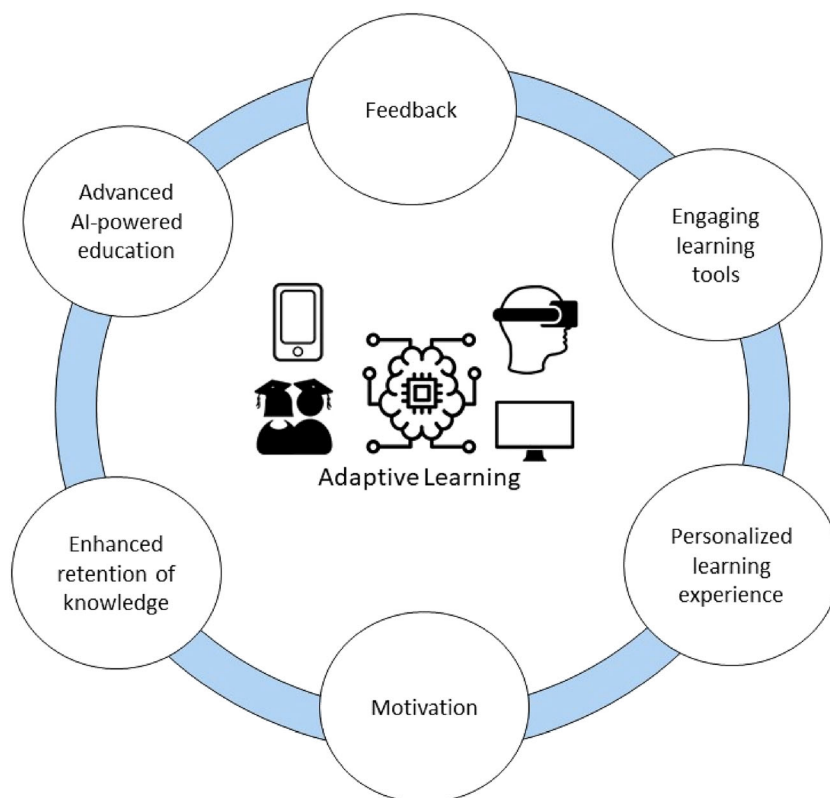
Figure 1 that follows offers a detailed visualization of the contemporary adaptive learning framework. This framework is supported by several technological pillars: personal computing, mobile computing, and spatial computing. It integrates advanced AI and augmented reality (AR) within educational approaches. Additionally, the figure illustrates engaging learning tools that are part of this framework. It also highlights personalized learning experiences designed for individual learners. Digitalized feedback mechanisms, aimed at enhancing the learning experience, are another key component depicted in the figure. Finally, the framework includes strategies for improved knowledge retention, which serve to increase learner motivation (see Figure 1).

Nevertheless, while adaptive learning technologies hold immense promise for revolutionizing education, many serious challenges remain. For instance, there are privacy concerns surrounding student data and the ethical implications of relying heavily on AI algorithms are areas that need careful consideration (Nguyen et al., 2023).

Our study aims at exploring the development of sustainable education with regard to the technological advancements and societal changes. In particular, it focuses on the impact of novel adaptive learning technologies and AI on education, especially with regard to the post-COVID sustainable education of the 21st century. Furthermore, the paper works towards critically analyzing the challenges and opportunities associated with adaptive learning technologies, concentrating on the ethical implications and privacy concerns related to use of AI algorithms. The ultimate goal is to provide a comprehensive understanding of the potential of adaptive learning in revolutionizing education while addressing the complexities and limitations associated with its implementation.

The novelty of this paper lies in its comprehensive analysis of the intersection between AI-driven adaptive learning technologies and sustainable education. While previous studies have explored adaptive learning and AI separately, this manuscript offers a unique perspective by examining how these technologies can be leveraged to not only

FIGURE 1 The framework for modern adaptive learning using personal computing, mobile computing, as well as spatial computing. Source: Own development.



enhance educational outcomes but also contribute to the broader goals of sustainability. By aligning educational practices with the United Nations' Sustainable Development Goals (SDGs), particularly in promoting quality education (SDG 4) and reducing inequalities (SDG 10), this study provides a fresh approach to understanding the role of technology in education.

The research topic tackled in this paper is important and timely due to its alignment with societal needs for sustainable education, its focus on cutting-edge technological advancements, its emphasis on individualized learning, and its scholarly rigor demonstrated through advanced analytical methods. The research addresses pressing educational challenges amplified by the digital surge accelerated by the COVID-19 pandemic, making it a crucial and timely contribution to the field of education. The paper is centered on the following three research questions (RQs):

1. **RQ1.** How have adaptive learning technologies and AI transformed contemporary education and contributed to the vision of sustainable education in the post-COVID era?
2. **RQ2.** What are the main challenges associated with implementing adaptive learning technologies?
3. **RQ3.** What key trends and patterns can be identified in the bibliometric analysis of the relevant publications on adaptive learning and AI, and

how can these findings be utilized for further research and policy development in sustainable adaptive education?

Through a bibliometric analysis and case studies, this paper identifies key trends, challenges, and opportunities in the implementation of adaptive learning technologies. It also offers practical insights into how these technologies can be effectively integrated into educational systems to foster sustainable development. What sets this study apart is its focus on the dual impact of AI-driven adaptive learning: improving personalized learning experiences while simultaneously advancing sustainability initiatives. This dual focus is particularly relevant in the current global context, where there is a growing recognition of the need for educational practices that are both effective and aligned with long-term sustainability goals.

The paper is structured in the following manner: Section 2 presents the structured literature review on adaptive learning and AI in enhancing and revolutionizing education. Section 3 explains the role of the adaptive learning in sustainable education of the 21st century. Section 4 contemplates over the challenges in implementing adaptive learning technologies using some examples and case studies. Section 5 features materials and methods describing the bibliometric analysis conducted and reported in this paper. Section 6 presents the results of the bibliometric analysis. Finally, the conclusions and implications section summarizes the results of this study, discusses its limitations, and offers the pathways for further research in this area.

2 | LITERATURE REVIEW

2.1 | Adaptive learning as a breakthrough in personalized education

Adaptive learning technologies and AI made a breakthrough in contemporary personalized education and contributed to the vision of sustainable education in the post-COVID era. In general terms, adaptive learning can be best understood as an intelligent system that leverages data analytics and machine learning algorithms to provide personalized instruction (Koutsantonis et al., 2022). Unlike traditional one-size-fits-all teaching methods, adaptive learning adapts the content, pace, and delivery of instruction based on an individual student's strengths, weaknesses, and learning preferences (Smyrнова-Trybulska et al., 2022).

Empirical evidence supports the claim that adaptive learning technologies significantly improve educational efficiency. Thence, the implementation of adaptive learning platforms might lead to higher pass rates and improved student retention compared to traditional teaching methods (Contrino et al., 2024; Demartini et al., 2024). For instance, one meta-analysis paper (García-Martínez et al., 2023) reviewed 25 studies on adaptive learning technologies and concluded that these technologies consistently enhance student engagement and performance across various educational contexts. All of this illustrates the substantial impact of adaptive learning technologies on educational efficiency, providing concrete evidence to support this claim.

In recent years, adaptive learning and personalized education encountered the growing interest of the general public and expert

alike. Figure 2 below shows the dynamics of the frequency of search requests (Interest over Time—search interest relative to the highest point on the chart for the given region and time and ranging from 0 (zero popularity on no data) to 100 (peak popularity)) of the search terms “personalized education” and “adaptive learning” for the last 20 years from 2004 until 2024.

Figure 2 was generated through an analysis of data retrieved from the Google Trends toolkit, a tool made available by the Google search engine. The toolkit identifies the changes in search queries of the main relevant concepts helping the researchers to assess the peak periods with the periods of the most significant changes in the searches on the Internet (Google Trends, 2023).

The customization used in adaptive learning ensures that learners receive targeted support and engage with materials at their optimal level of challenge. At the core of adaptive learning is its data-driven nature. Learning platforms equipped with adaptive technologies collect vast amounts of data about students' performance, interactions, and progress throughout their educational journey (Kamalov et al., 2023). These platforms use this data to create detailed learner profiles that capture each student's knowledge gaps, misconceptions, preferred learning styles, and areas of proficiency. With these learner profiles in hand, adaptive systems employ sophisticated algorithms to analyze the data and make informed decisions about what content a student should learn next or how it should be delivered (Wolff et al., 2023). For instance, if a student struggles with fractions but excels at geometry concepts, the system will identify this discrepancy through data analysis and adjust the instructional materials accordingly by providing additional practice or alternative explanations

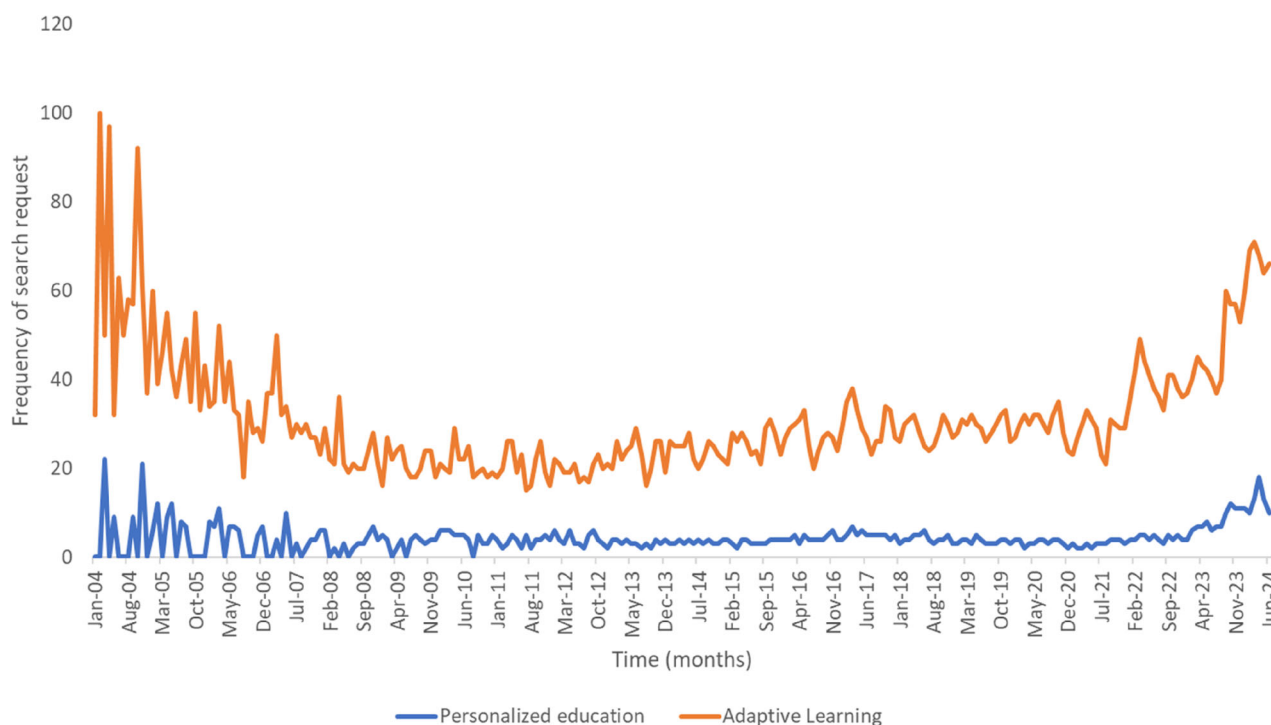


FIGURE 2 Dynamics of frequency of search of the terms of “personalized education” and “adaptive learning” (2004–2024). Source: Own results based on Google Trends.

specifically targeting fractions. Moreover, adaptive learning does not limit itself to adjusting content but also considers other crucial factors such as pacing and feedback provision. The system monitors students' progress in real-time and adapts to the pace at which new material is introduced based on their demonstrated mastery level. This ensures that learners are neither overwhelmed nor bored by content that is too easy or too challenging for them (Avram & Pop, 2023; Wolff et al., 2023). Additionally, adaptive systems provide immediate feedback throughout the learning process. This timely feedback not only enhances students' understanding but also promotes metacognition and self-regulation skills as they actively reflect on their progress (Garrison, 2022).

Thence, the benefits of adaptive learning extend beyond individual learners to educators as well. Adaptive systems use machine learning to assess the learning patterns and study progress of the students in real time and to report them to tutors who can, in turn, quickly adapt their teaching strategies, intervene when needed or adjust their lectures according to the students' progress and understanding of the material.

2.2 | AI in revolutionizing education: The power of machine learning

One significant advantage AI brings to education is its ability to analyze large amounts of data quickly. By collecting vast quantities of student information, such as performance data, preferences, and learning patterns, AI algorithms can create individualized learning paths for each student (Shemshack et al., 2021). Adaptive learning platforms leverage this capability by continuously monitoring students' progress and adapting instructional content accordingly. This personalized approach ensures that students receive targeted support where they need it most (Muñoz et al., 2022).

It is important to differentiate between various types of adaptive learning technologies, as they offer different functionalities and cater to diverse educational needs. Content-based adaptive systems are designed to personalize the instructional content delivered to students, adjusting based on their progress and learning preferences. These systems are particularly effective in subjects where the learning material is cumulative, such as mathematics or language learning. Assessment-based adaptive systems, on the other hand, focus on tailoring assessments and providing personalized feedback, dynamically adjusting the difficulty level of tasks to match the student's performance. Finally, integrated adaptive systems combine both content delivery and assessment features, offering a comprehensive solution that supports students through personalized learning paths, continuous assessment, and targeted interventions. Understanding these distinctions is essential for educators and institutions when selecting the most suitable adaptive learning technologies for their specific contexts.

Several case studies or examples can be mentioned here to demonstrate the successful application of AI in adaptive learning environments: In the United States, Arizona State University (ASU) partnered

with Knewton to implement an adaptive learning platform in college-level math courses. The AI-driven platform analyzed individual student data to identify areas of struggle and provided personalized recommendations. As a result, students who used the platform achieved higher pass rates compared to those who did not, illustrating the effectiveness of AI in enhancing academic performance (Maxfiel, 2023).

Another example is DreamBox Learning, an adaptive math program used in elementary schools, leverages AI to track students' progress in real-time. By identifying knowledge gaps and adjusting lessons accordingly, the program has significantly improved math proficiency among its users (Foster, 2023). A study conducted by Harvard University found that students who used DreamBox regularly demonstrated substantial gains in math skills compared to their peers using traditional methods (Barbieri et al., 2020).

Yet another real-world example is Carnegie Learning platform that utilized AI-powered adaptive technology to enhance its middle school math curriculum. The platform collected data on student performance and generated personalized assignments tailored to each student's strengths and weaknesses. Schools implementing Carnegie Learning's program reported improved test scores and greater student engagement (Wested., 2017). In addition to these examples, many universities have embraced adaptive learning platforms like Smart Sparrow and CogBooks to enhance teaching effectiveness across diverse subjects such as science, engineering, and humanities (Li et al., 2021; Putra et al., 2022). All of these case studies discussed above clearly demonstrate the transformative impact of adaptive learning on education.

Machine learning algorithms also facilitate real-time feedback for students. Traditional assessment methods often provide feedback after completion or during specific intervals, limiting immediate corrections or improvements (Sharma et al., 2020). However, with AI-driven systems, learners receive instant feedback on their work through automated grading systems or virtual tutors capable of guiding at any time. This immediate response helps students identify their mistakes promptly and make necessary adjustments while still engaging with the topic at hand (Celik et al., 2022).

With regard to the above, the relationship between adaptive learning technologies and traditional assessment methods appears to be a critical area of consideration in education as well in the public sector (Marjerison & Gatto, 2024). While traditional assessments, such as standardized tests and exams, provide summative evaluations at specific points in time, adaptive learning technologies offer continuous, formative assessments that adjust to the learner's needs in real time. This ongoing evaluation process allows for more personalized feedback and supports the individual learning journey. However, integrating these adaptive approaches with traditional assessment methods poses challenges, such as aligning real-time feedback with standardized criteria and ensuring consistency in measuring student progress (Mirata et al., 2020). A balanced approach that combines the strengths of both traditional and adaptive assessments can enhance the overall evaluation process, providing educators with a more comprehensive understanding of student learning outcomes which is

especially relevant in the era of upcoming Industry 5.0 that offers innovative new prospects leading to the production of more environmentally friendly projects, services, and products (Baig & Yadegaridehkordi, 2024).

Moreover, machine learning algorithms in education revolutionized the way feedback is delivered, enabling students to receive real-time responses to their work. This immediacy in feedback not only assists learners in promptly identifying and correcting their mistakes but also ensures continuous engagement and adjustment within the learning process, thereby significantly enhancing learning outcomes.

Another aspect where machine learning enhances education is in creating immersive and interactive experiences for learners. Virtual reality (VR) and AR technologies powered by AI enable students to explore subjects like history or science through simulated environments that would otherwise be inaccessible or dangerous in real life (Scavarelli et al., 2021). For instance, VR can transport history students to ancient civilizations or allow biology classes to dissect virtual organisms without any ethical concerns. Moreover, leveraging the power of machine learning enables educators to predict future student performance accurately. By analyzing patterns from past data combined with contextual factors like attendance records or engagement levels, algorithms can forecast potential challenges a student might face in their academic journey. With this information, teachers can intervene early, providing timely support, and guidance to help struggling students before they fall behind (Atalla et al., 2023; Cheng et al., 2023).

In addition, the role of AI in revolutionizing education also extends beyond the classroom. It can be used for routine administrative tasks such as preparing study schedules and plans, progress monitoring, or grading, and marking. However, it cannot completely replace humans in any of those processes. The human touch remains vital for fostering creativity, critical thinking skills, and social-emotional development—areas where machines currently lack proficiency (Heim et al., 2023).

2.3 | Intelligent tutoring systems: Enhancing student engagement and performance

With the rapid rise of adaptive learning and AI, education has witnessed a significant transformation. Intelligent tutoring systems (ITS) have emerged as powerful tools that not only enhance student engagement but also improve academic performance (Guo et al., 2021). ITS addresses this issue by offering individualized instruction that aligns with students' knowledge gaps, learning styles, and pace. By employing advanced machine learning techniques, intelligent tutoring systems analyze student data in real time. They monitor their progress, identify areas of weakness, and provide targeted feedback for improvement (Li & Xia, 2020; Mahmood et al., 2022).

This personalized approach ensures that students receive tailored support precisely where they need it most. As a result, learners feel more motivated and engaged in their studies as they witness tangible progress. Furthermore, intelligent tutoring systems foster

independent learning by encouraging students to take ownership of their educational journey. Through adaptive algorithms, ITS guides learners through challenging concepts at an appropriate level of difficulty while gradually building their skills (Benvenuti et al., 2023). This approach promotes self-directed learning habits that are crucial for lifelong success. Beyond enhancing engagement levels, intelligent tutoring systems have also demonstrated significant improvements in academic performance across various subjects (Chen et al., 2020). Many studies demonstrated that students using ITS outperform those relying solely on traditional teaching methods or even computer-based instruction without adaptability (Muca, Cavallini, et al., 2023; Rejmi & Rasli, 2022). Thence, intelligent tutoring systems are revolutionizing education by providing personalized instruction tailored to individual student needs. By leveraging AI algorithms and real-time data analysis, these systems not only enhance engagement levels but also boost academic performance across a wide range of subjects.

2.4 | Novel educational solutions: Transforming traditional teaching methods

In the past few decades, AI and adaptive learning took education by surprise helping to reinvent the traditional methods of teaching for both lecturers and students and providing personalized educational experiences in real time. Traditional teaching methods that used to rely on a one-size-fits-all approach, where educators deliver content at a fixed pace without considering variations in student abilities or interests, suddenly became obsolete (Alamri et al., 2021).

However, with AI-powered platforms, students can receive tailored instruction that caters to their unique strengths and weaknesses. By analyzing vast amounts of data, such systems can identify individual learning patterns and adjust their curriculum accordingly (Tapalova & Zhiyenbayeva, 2022). This personalized approach not only enhances student engagement but also improves overall learning outcomes. Moreover, AI-driven educational solutions offer real-time feedback and assessment capabilities that go beyond what traditional teaching methods can provide (Wongvorachan et al., 2022).

Through continuous monitoring of students' progress, these platforms can identify areas where additional support is needed and offer targeted interventions to address those gaps. This instant feedback loop empowers students to take ownership of their learning journey by enabling them to track their progress and make informed decisions about areas requiring further attention (Meng, 2023). Additionally, AI-powered platforms facilitate collaborative learning experiences by connecting students with peers who share similar interests or academic goals. By creating virtual communities, these solutions foster a sense of belonging and encourage knowledge sharing among learners from diverse backgrounds (He et al., 2023). This collaborative environment promotes critical thinking skills, problem-solving abilities, and communication proficiency—essential competencies for success in the digital age (Minyar-Beloroucheva & Sergienko, 2022). With all that, AI-driven educational solutions are transforming traditional teaching methods by providing personalized instruction tailored to individual

student needs. With adaptive learning algorithms that analyze vast amounts of data in real time, these platforms offer customized curricula while facilitating collaborative experiences among learners.

2.5 | Adaptive assessment systems: Redefining testing and evaluation

The integration of adaptive assessment systems in education has revolutionized the field of testing and evaluation. These innovative systems utilize AI algorithms to dynamically adjust the difficulty level and content of assessments based on a student's performance. This personalized approach to testing offers numerous benefits that traditional standardized tests simply cannot match (Maghsudi et al., 2021).

Unlike traditional exams, where students have to wait for days or even weeks for their results, adaptive assessments generate instant feedback. This real-time evaluation allows students to identify their strengths and weaknesses promptly, enabling them to focus on areas that require improvement and reinforcing their learning experience (Chen, 2022). Moreover, adaptive assessment systems offer a more accurate representation of a student's knowledge and skills compared to traditional tests. These systems adapt the difficulty level of questions based on the student's responses, ensuring that each question is appropriately challenging. By tailoring the assessment to an individual's abilities, these systems eliminate the potential biases associated with one-size-fits-all exams and provide a more accurate measure of a student's true capabilities (Owan et al., 2023). Additionally, adaptive assessments encourage active engagement and motivation among students.

As each question is tailored to their specific abilities, students feel challenged but not overwhelmed, fostering a sense of achievement and motivation. Furthermore, adaptive assessment systems enable educators to gather valuable data about individual student performance as well as overall class progress (Zainuddin et al., 2020). The AI algorithms collect comprehensive data on each student's strengths, weaknesses, learning patterns, and progress over time. This data-driven approach allows teachers to gain insight into individual needs and make informed decisions regarding instructional strategies or interventions tailored specifically for each student (Glover et al., 2023).

2.6 | Adaptive curriculum design: Tailoring education to individual needs

Adaptive learning and AI offer immense potentials in adjusting education to individual needs. The one-fits-all approach gives way to the AI-powered curriculum design tailored to each student in accordance with her or his abilities and interests. (Jaiswal & Arun, 2021). AI allows to monitor student's learning pace in real time while identifying her or his strengths and weaknesses which generates useful feedback for the teachers and instructors who can adjust their methods and approaches.

With AI-powered curriculum design, personalized learning becomes a reality. Adaptive learning algorithms can customize educational content based on an individual student's proficiency level and learning style (Kabudi et al., 2021; Martin et al., 2020).

In addition, the system can help the student to grasp the new ideas and concepts by offering alternative explanations. Conversely, if a student demonstrates advanced understanding in certain areas, the system can offer more challenging material or opportunities for further exploration (Kasneji et al., 2023). Furthermore, AI-powered systems facilitate continuous assessment throughout the learning process. Instead of relying solely on standardized tests at predetermined intervals, these systems continuously evaluate students' knowledge and skills through interactive exercises and quizzes (Vesin et al., 2022).

By providing immediate feedback and adapting the difficulty level based on performance, AI-powered curriculum design fosters an environment where mistakes are seen as opportunities for growth rather than failures. Such curriculum design represents a significant step towards tailoring education to individual needs (Wiggins et al., 2020).

Table 1 that follows summarizes the main provisions and discussions stemming from the literature review presented in this section.

2.7 | Limitations of AI in education

Nevertheless, it has to be remembered that while AI has the potential to revolutionize education by providing personalized learning experiences and improving efficiency, it is also important to recognize the limitations of this technology as highlighted in the existing literature (Akgun & Greenhow, 2022; Zhai et al., 2021).

One of the significant concerns with AI in education is the potential for algorithmic bias. AI systems that are trained on biased datasets can inadvertently perpetuate or even exacerbate inequalities. For instance, some studies demonstrated that AI algorithms may disadvantage students from underrepresented groups by providing less accurate or biased feedback, thus impacting their learning outcomes (Baker & Hawn, 2022; Von Winckelmann, 2023).

In addition, the extensive data collection required for AI-driven educational tools raises serious privacy concerns. There is a risk of data breaches, unauthorized access, and misuse of sensitive student information. The literature emphasizes the need for robust data protection measures and clear policies to ensure that student privacy is safeguarded (Berendt et al., 2020).

Furthermore, there is also the risk of over-reliance on AI technologies, where educators and students may depend too heavily on automated systems (Zhai et al., 2024). This reliance can lead to a diminished role for critical thinking and human judgment, as well as a potential decrease in the development of problem-solving skills that are fostered through traditional educational practices.

Another limitation noted in the literature is the potential reduction in meaningful human interactions between teachers and students (Markauskaite et al., 2022). AI systems, while efficient, cannot

TABLE 1 Key elements of AI, novel technologies, and approaches in adaptive learning.

Key elements	Characteristics and main provisions
Adaptive learning	- Uses data analytics and AI to offer personalized instruction - Content, pace, and delivery adapt based on students' strengths, weaknesses, and preferences. - Data-driven nature collects student performance data. - Creates learner profiles for customized learning paths. - Algorithms analyze data for content and delivery decisions. - Adjusts pacing and feedback provision. - Enhances metacognition and self-regulation skills
AI in education	- Analyzes large student data for individualized learning paths. - Offers continuous progress monitoring and content adaptation. - Real-time feedback through automated grading and virtual tutors. - Immersive experiences using VR and AR. - Predicts future student performance for early interventions. - AI assists in administrative tasks, complementing the human touch.
Intelligent tutoring systems	- Offer individualized instruction. - Real-time data analysis for progress monitoring. - Targeted feedback to address weaknesses. - Promote engagement and self-directed learning. - Improve academic performance.
AI-driven novel educational solutions	- Transforms traditional teaching with personalized experiences. - Tailored instruction based on student strengths and weaknesses. - Real-time feedback and assessment. - Continuous monitoring for support and interventions. - Facilitates collaborative learning. - Enhances critical thinking and problem-solving.
Adaptive assessment systems	- Adjust difficulty based on student performance. - Real-time feedback for instant improvement. - Accurate representation of knowledge and skills. - Encourage active engagement and motivation. - Provide valuable data for educators.
Adaptive curriculum design	- Tailors curriculum to individual student needs. - Monitors progress for insights into strengths and weaknesses. - Personalized content based on proficiency and learning style. - Continuous assessment and immediate feedback. - Fosters growth mindset.

Abbreviation: AI, artificial intelligence.

Source: Own results.

replicate the social and emotional learning that occurs through human connections, which are vital for a well-rounded education.

Finally, it is apparent that implementing AI in diverse educational settings presents significant challenges. In resource-constrained environments, the lack of infrastructure, training, and ongoing support can significantly limit the effective use of AI. Additionally, the high costs associated with acquiring and maintaining AI technologies can be prohibitive for many educational institutions.

By acknowledging all these limitations, one can adopt a more cautious and balanced approach to integrating AI into education, ensuring that its deployment is done in a way that maximizes benefits while mitigating potential downsides.

3 | ADAPTIVE LEARNING IN SUSTAINABLE EDUCATION

Nowadays, the sustainable future represents one of the main priorities for our society and the sustainable education is becoming an

important tool how to prepare the next generations to come for face complex environmental and social challenges. Adaptive learning, a dynamic approach that tailors educational experiences to individual needs, holds immense potential in achieving this goal (Gorina et al., 2023; Kadaruddin, 2023).

In particular, adaptive learning technologies can be tailored to address specific learning disabilities or special educational needs, making education more inclusive. These technologies can be customized to offer personalized learning experiences for students with disabilities such as dyslexia, ADHD, autism spectrum disorders, and sensory impairments (Barua et al., 2022; Kem, 2022). For example, adaptive platforms can adjust the difficulty level, provide alternative explanations, or present information in various formats (visual, auditory, tactile) to suit the learning preferences and challenges of individual students. Additionally, these technologies can offer immediate feedback and reinforcement, which is particularly beneficial for students who require more frequent support. By incorporating features that specifically cater to the needs of students with learning disabilities, adaptive learning technologies contribute to creating inclusive

educational environments where all students have the opportunity to succeed.

Furthermore, adaptive learning can adapt content and delivery methods based on learner progress and preferences. It appears that adaptive learning technologies and AI can significantly contribute to the vision of sustainable education in our post-COVID era with benefits in fostering student engagement, enhancing educational outcomes, and promoting lifelong learning for a more sustainable future (Yuan et al., 2021).

To better understand the impact of our study on the SDGs, it needs to be explored how adaptive learning technologies and AI contribute to specific SDGs in various contexts (Makurumidze et al., 2024). Primarily, SDG 4 (Quality Education) is directly influenced by adaptive learning technologies that ensure inclusive and equitable quality education by providing personalized learning experiences (Zhukova et al., 2022). These technologies address diverse learning needs and styles, reduce dropout rates, and improve learning outcomes, ensuring that all students, regardless of their background or abilities, have access to high-quality education. The example of Arizona State University's implementation of Knewton's adaptive learning platform described above demonstrated the potential of these technologies to enhance educational quality.

Furthermore, adaptive learning technologies support SDG 8 (Decent Work and Economic Growth) by enhancing skill development and aligning educational content with the demands of the modern workforce. By equipping students with relevant skills and knowledge, these technologies improve employability and productivity, thus contributing to economic growth. DreamBox Learning, an

adaptive math program also described above, has been shown to significantly improve math proficiency among elementary students, laying a strong foundation for their future success in the workforce.

Additionally, adaptive learning technologies contribute to SDG 10 (Reduced Inequalities) by bridging educational gaps between different socio-economic groups. Personalized learning experiences ensure that all students, including those from marginalized communities, receive the necessary support to succeed academically (Opesemowo & Adekomaya, 2024). The use of AI-driven adaptive learning platforms in diverse educational settings has shown promising results in reducing achievement gaps between students from different socio-economic backgrounds.

Finally, adaptive learning technologies promote SDG 13 (Climate Action) by integrating environmental education into personalized learning paths (Apoki et al., 2022). This fosters a generation of environmentally conscious individuals who are better equipped to understand and address climate change and sustainability issues. Adaptive learning platforms can include modules on climate change and sustainability, encouraging students to engage with and understand these critical global issues from a young age.

Figure 3 that follows shows the dynamics of the frequency of search requests using the search items “adaptive learning” and “sustainable education” (Figure 3). One can see that the search frequency for both terms is quite similar and both terms are often being searched simultaneously after 2013–2014. The parallel increase in search frequency for “adaptive learning” and “sustainable education” post-2014 may reflect a broader shift in educational priorities towards more personalized, technologically integrated, and long-term

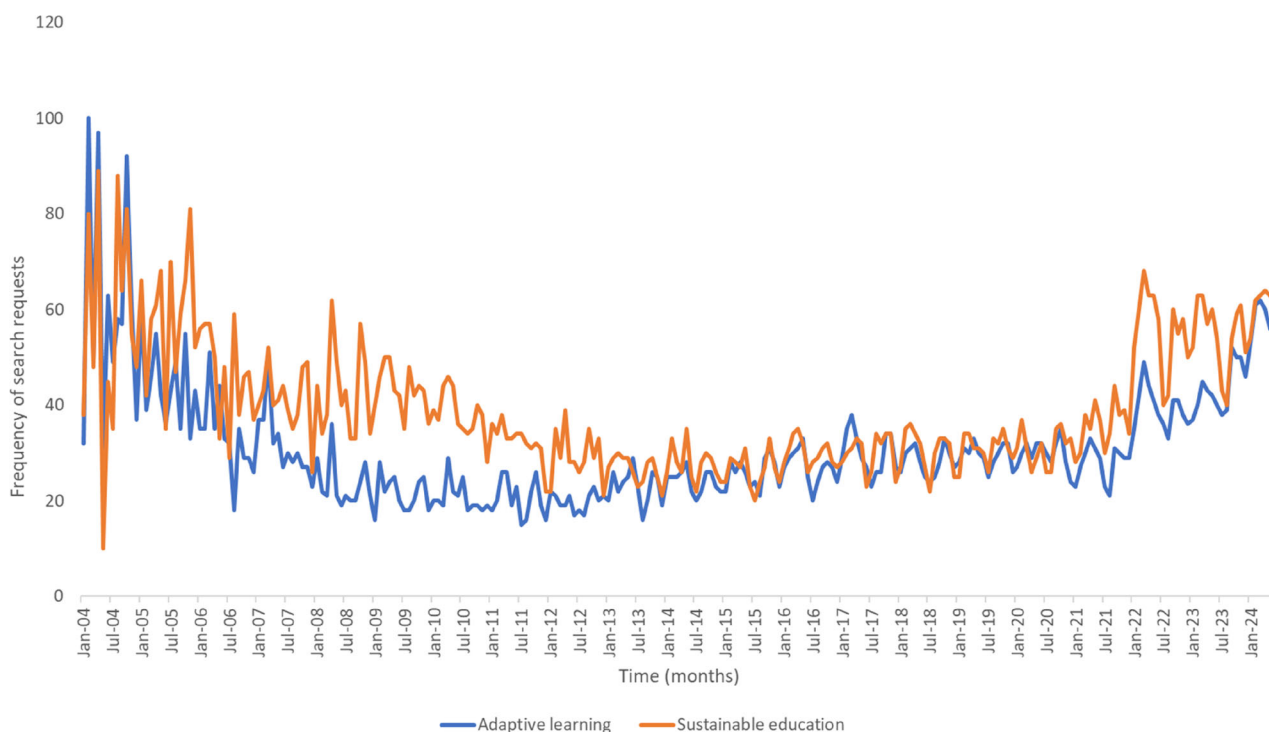


FIGURE 3 Dynamics of frequency of search of the terms of “adaptive learning” and “sustainable education” (2004–2024). Source: Own results based on Google Trends.

sustainable educational practices. This correlation can be attributed to several factors: (i) adaptive learning aligns with sustainable education, which aims not only for environmental sustainability but also for sustainable, long-term learning strategies that benefit learners over time; (ii) technologies used in adaptive learning align well with the principles of sustainable education, which often include the integration of technological advancements to create more efficient and enduring educational practices; (iii) adaptive learning and sustainable education align with global educational goals, such as the UN SDG 4; (iv) there has been a noticeable increase in research and policy development around both adaptive learning and sustainable education; (v) the education technology market's growth likely corresponds with an increased interest in sustainable education, as both areas benefit from technological innovation.

Adaptive learning holds immense potential to revolutionize sustainable education by catering to the individual needs of learners. Firstly, it allows for personalized instruction, enabling students to learn at their own pace and style. By adapting the content, delivery, and assessment methods to suit each student's unique requirements, adaptive learning ensures a more engaging and effective learning experience (Marougkas et al., 2023). Additionally, this approach promotes self-directed learning as students take ownership of their education by setting goals and monitoring progress. Moreover, adaptive learning systems collect vast amounts of data on student performance, allowing educators to identify knowledge gaps and tailor interventions accordingly (Currie et al., 2022). This data-driven approach empowers teachers to provide targeted support while optimizing resource allocation for a more efficient educational system that aligns with SDGs (Romanova & Kuzmin, 2021).

The implementation of adaptive learning in sustainable education brings forth a transformative approach to teaching and learning. By integrating technology, data analytics, and personalized instruction, adaptive learning empowers educators to tailor educational experiences according to individual needs, preferences, and learning styles. Through the utilization of intelligent algorithms, adaptive learning platforms continuously assess students' progress and adapt instructional content accordingly. This personalized approach fosters greater engagement and motivation among students, leading to improved academic outcomes (Aulakh et al., 2023).

Additionally, by leveraging digital resources and reducing bureaucracy and paperwork, adaptive learning contributes to a more sustainable educational environment. The implementation of adaptive learning also facilitates the development of lifelong learners who are equipped with the necessary skills to thrive in an ever-changing world while promoting sustainability as an integral part of their education journey (Holman & Švejdarová, 2023).

Adopting adaptive learning for sustainable education presents a unique set of challenges that require innovative solutions. One major obstacle is the limited access to technology and internet connectivity in many developing regions. To overcome this, offline adaptive learning platforms can be developed, allowing students to access educational content without an internet connection (Tzenios, 2020). Another challenge is the lack of trained educators who can effectively

implement adaptive learning strategies (Adarkwah, 2021). Addressing this issue requires comprehensive training programs and support for teachers to enhance their digital literacy skills and enable them to leverage adaptive learning tools effectively. Additionally, the high cost of implementing adaptive learning systems can hinder widespread adoption. Collaborative efforts between governments, educational institutions, and technology providers are necessary to develop affordable solutions that make adaptive learning accessible to all learners, ensuring sustainable education for future generations (Kant et al., 2021).

As technology continues to advance, the future prospects of adaptive learning in sustainable education are promising. With the ability to personalize instruction and tailor content to individual learners, adaptive learning has the potential to revolutionize education (Muca, Buonaiuto, et al., 2023). By leveraging data analytics and AI, educators can gain valuable insights into students' strengths and weaknesses, facilitating targeted interventions and personalized support (Simionescu et al., 2022). Thence, adaptive learning can promote sustainability by reducing paper waste through digital resources and fostering eco-friendly practices as well as integration for students regardless of their physical location or background.

4 | CHALLENGES IN IMPLEMENTING ADAPTIVE LEARNING TECHNOLOGIES

The integration of adaptive learning technologies into education has brought significant challenges apart from otherwise significant advancements. As educational institutions strive to revolutionize the traditional classroom experience, they must address several hurdles to ensure the successful implementation of adaptive learning technologies (Sá & Serpa, 2020). However, there is always the need for continuous updates and maintenance of adaptive learning technologies to ensure they remain effective. These systems rely on sophisticated algorithms and large datasets that must be regularly updated to stay aligned with evolving educational standards, technological advancements, and changing student needs. Regular updates are essential for incorporating new content, improving personalization, fixing technical issues, and ensuring data security. Without ongoing maintenance and updates, adaptive learning technologies risk becoming outdated and less effective, which could diminish their value in educational settings. Institutions must therefore plan for continuous support and upgrading of these systems as part of their long-term strategy for successful implementation (Lee & Lee, 2024).

Moreover, one should not forget about the variability in technology access among different population (Major et al., 2021). The integration of adaptive learning technologies and AI in education must account for disparities in technology access to ensure equitable educational opportunities. In many regions, especially in developing countries, limited access to reliable internet connectivity, digital devices, and technical support poses significant barriers to the effective implementation of these technologies. Addressing these disparities is

crucial for realizing the full potential of adaptive learning and AI in promoting inclusive and equitable education.

Another critical concern is the potential for adaptive learning technologies to contribute to widening the digital divide. While these technologies can significantly enhance educational outcomes, they may also exacerbate existing inequalities if access to them is not distributed equitably (Dumont & Ready, 2023). Students from disadvantaged backgrounds, rural areas, or lower-income families may lack the devices, internet connectivity, or digital literacy skills needed to effectively engage with adaptive learning tools. This lack of access could further marginalize these students, widening the gap between them and their peers who have better access to technology. To prevent this outcome, it is essential to implement strategies that promote equitable access to adaptive learning technologies, including infrastructure development, affordable devices, and digital literacy programs. By addressing these issues, one can ensure that adaptive learning technologies help close the digital divide rather than widen it.

In addition, there are other several challenges to be considered. First, there is resistance to change from educators who are accustomed to conventional teaching methods and may be skeptical about the effectiveness of new technologies (Núñez-Canal et al., 2022). This resistance can be due to a lack of understanding of how these technologies work or concerns about the additional time and effort required to integrate them into existing curricula. Second, the infrastructure in traditional schools may not support advanced technological tools, requiring significant investment in hardware, software, and reliable internet connectivity (e.g., in places such as some African countries) (Maphosa, 2021). Third, there are cultural and institutional barriers, where established educational norms and policies may not align with the flexible, personalized approaches offered by adaptive learning technologies (Ayeti et al., 2024). Addressing these challenges requires comprehensive training for educators, upgrades to infrastructure, and policy adjustments to facilitate the seamless integration of adaptive learning technologies.

Another critical issue is the potential for AI bias in educational tools which arises when algorithms produce skewed results that disadvantage certain student groups, often due to biased training data or flawed algorithm design (Ferrara, 2023). For instance, an AI system trained predominantly on data from a specific demographic may not perform equally well for students from diverse backgrounds, leading to unequal educational opportunities and outcomes. To mitigate AI bias, it is essential to ensure diverse and representative data sets, involve diverse teams in the development process, and continuously monitor and audit AI systems for biases. By implementing these measures, we can help create more fair and inclusive educational environments that benefit all students.

Resistance from educators and institutions is an additional common challenge in the adoption of adaptive learning technologies. Educators may resist these technologies due to a lack of familiarity, fear of obsolescence, or concerns about the time and effort required for training. Skepticism about the effectiveness of adaptive learning tools and the potential reduction in personal interaction with students can also contribute to resistance (Alzahrani et al., 2023). From an

institutional perspective, concerns about the financial investment, potential disruption to established processes, and uncertainty about long-term benefits can hinder the adoption of these technologies. To address these concerns, it is crucial to provide comprehensive training, clearly communicate the benefits, and involve educators in the decision-making process. By addressing resistance proactively, institutions can pave the way for a smoother and more successful integration of adaptive learning technologies.

Furthermore, cultural differences can also significantly impact the acceptance and effectiveness of adaptive learning technologies (Essa et al., 2023). Cultural attitudes towards technology, education, and authority can shape how students, educators, and institutions perceive and interact with these tools. For instance, in cultures that emphasize teacher-centered instruction, there may be resistance to the more autonomous, student-centered learning models promoted by adaptive technologies. Furthermore, cultural variations in communication styles, learning preferences, and the role of feedback can affect how effectively students engage with these technologies. To ensure the successful implementation of adaptive learning systems, it is crucial to consider these cultural factors and adapt the technologies to align with local educational practices and values.

In order to mitigate these challenges, it is essential to adopt strategies that provide offline adaptive learning solutions, ensuring that students can access educational content without continuous internet connectivity. Additionally, government and private sector partnerships can play a pivotal role in enhancing infrastructure, providing affordable digital devices, and offering training programs for educators and students to improve digital literacy. By focusing on these measures, we can bridge the digital divide and ensure that the benefits of adaptive learning technologies are accessible to all students, regardless of their socio-economic background or geographic location.

One of the foremost challenges is resistance to change. Educational systems have been structured around traditional teaching methods for centuries, and many educators may be hesitant to embrace new technologies. Convincing teachers and administrators that adaptive learning can enhance student outcomes requires a comprehensive approach that emphasizes its benefits, such as personalized instruction and improved student engagement. Another significant challenge lies in the availability and quality of digital content. Adaptive learning relies heavily on content that can be tailored to each student's needs and preferences (Raj & Renumol, 2022). However, creating such content demands considerable resources, expertise, and time. To overcome this obstacle, collaborations between content developers and educators should be fostered to ensure the creation of high-quality digital resources that align with curriculum standards (Sanguliano Intra et al., 2023).

Additionally, one more important consideration to be considered is the potential dependency on technology that adaptive learning might create among students (Wang et al., 2023). As students become accustomed to the personalized guidance and instant feedback provided by these systems, they may develop a reliance on technology that reduces their ability to engage in self-directed learning and independent problem-solving. This dependency could undermine the

development of critical thinking skills, as students may struggle to manage their learning processes without the support of adaptive technologies. To mitigate this risk, educators should implement strategies that promote a balanced approach, encouraging the development of autonomous learning skills alongside the use of adaptive technologies. This could include integrating traditional learning methods, fostering activities that require independent thought, and providing opportunities for students to practice problem-solving without technological assistance.

In addition to these challenges, the cost implications of implementing adaptive learning technologies are a significant consideration. The initial investment required for software licenses, hardware upgrades, and the installation of necessary infrastructure can be substantial. Ongoing costs, including software maintenance, updates, and technical support, also need to be factored into the budget (Walker et al., 2023). Moreover, training and professional development for educators represent an essential but often overlooked cost, as teachers must be adequately prepared to use adaptive learning tools effectively. The financial burden can vary depending on the size of the educational institution, the extent of existing technological infrastructure, and the specific adaptive learning solutions chosen. A comprehensive cost-benefit analysis is critical to determine whether the educational benefits of these technologies justify the investment and to ensure their sustainable implementation across diverse educational contexts.

Furthermore, there is a need for robust infrastructure and technology support in educational institutions. Implementing adaptive learning technologies requires reliable internet connectivity, adequate hardware devices like computers or tablets, software compatibility across different platforms or operating systems, as well as technical support for troubleshooting issues promptly (Kesavan et al., 2022). Data privacy concerns also pose a significant challenge in implementing adaptive learning technologies. The collection of personal data from students raises ethical questions regarding how it will be used or shared with third parties (Jones, 2019; Lutz & Newlands, 2021). Establishing clear policies on data privacy protection and compliance with relevant regulations is essential to address these concerns effectively. All in all, overcoming challenges associated with resistance to change among educators, availability of quality digital content, infrastructure limitations, and data privacy concerns are crucial for their successful implementation (Ahmad, Mohd Noor, et al., 2023; Almaiah et al., 2020).

Yet another crucial element in overcoming these obstacles is the role of teacher training and professional development. Educators must be equipped with the necessary skills to effectively use adaptive learning technologies in their classrooms (Li et al., 2023). This includes understanding how to personalize learning experiences using these tools, interpreting data-driven insights to inform instruction, and continuously adapting their teaching strategies. Professional development programs should be designed to provide ongoing support and training, ensuring that educators remain confident and competent in using adaptive technologies. Without adequate training, the full potential of

these technologies may not be realized, limiting their impact on educational outcomes.

The ability of AI and adaptive learning to personalize the learning experience based on individual needs and abilities has garnered widespread recognition and adoption across various educational institutions. One can explore some real-world case studies highlighting the success of adaptive learning in transforming education (Mian et al., 2020).

By employing the AI technology to deliver personalized instruction, educators can effectively address individual student needs while fostering a deeper understanding of complex concepts across various academic disciplines.

5 | MATERIALS AND METHODS

In this section of our paper, we apply the bibliometric approach that is based on the software tool VOSviewer (Version: 1.6.18) to delineate the foundational contributions within the topic of adaptive learning and AI within the period spanning 1990 to 2023, utilizing the Web of Science (WoS) database. Bibliometrics represents a metrological approach amalgamating mathematical, statistical, and bibliographic techniques. It entails a quantitative assessment and categorization of an extensive body of scholarly work based on published texts (Adisa et al., 2002; Tao et al., 2021). Today, bibliometrics constitutes an important part of data science and is often used for the quantitative analysis of the research literature using such methods as cluster analysis and mapping (Sahoo, 2022; Strielkowski et al., 2022).

5.1 | Data and algorithm selection

We employed the VOSviewer software to perform the statistical analyses on the publications indexed in WoS featuring such information as countries, authors, abstracts, and keywords to analyze the major research forces in AI and adaptive learning with the help of assessing the co-occurrences and keywords' cluster analyses. Table 2 that follows provides the summary of data and the algorithm for the data selection.

In addition, Figure 4 outlines the methods used in our study describing the study selection criteria and the screening process that included searching within the title, abstract and not just keywords (which is important, since some technical papers without focusing on learning, may use the keywords but might not align with the perspective of this review).

5.2 | Sample of publications

We have chosen the WoS database and conducted a search using the terms "adaptive learning" and "artificial intelligence" which retrieved a total number of 3518 indexed publications (2501 articles and 762 proceeding papers among them) (see Table 2 above).

Figure 5 that follows depicts the trend of the publications on adaptive learning over time using the sample of publications from WoS database (1990–2023). We have deliberately excluded the year of 2024 in order not to bias the growing dynamics because the publication count is still ongoing.

Looking at the trends shown in Figure 5, it becomes apparent that the main spike in the publications on adaptive learning and AI occurred around 2018–2019 which was further enhanced by the COVID-19 pandemic in 2020 and the rapidly growing interest and demand for everything digital applied to education that was caused by the “digital surge” that the pandemic caused.

TABLE 2 Summary of data and data selection algorithm.

Category	Specific criteria
Reference and citation database	Web of Science
Citation indices	SCI-Expanded, SSCI
Time period	1990–2024
Language	“English”
Keywords	“adaptive learning” AND “artificial intelligence”
Document types:	
Articles	2501
Proceeding papers	762
Others	188
Sample size	$N = 3518$

Source: Own results.

6 | EMPIRICAL ANALYSIS: BIBLIOMETRIC APPROACH

6.1 | Data analysis: Journals, institutions, and publications

We commence our bibliographic analysis with the scrutiny of data represented by top journals publishing research on adaptive learning and AI between 1990 and 2024, as well as top prolific institutions working on these subjects. In addition, we present the most cited publications on adaptive learning and AI in education and e-learning.

In order to systematically assess the effectiveness of adaptive learning strategies, it is crucial to adopt a clear framework that encompasses multiple dimensions of evaluation. This framework should include:

1. Measure student engagement by assessing the level of student interaction with the adaptive learning platform, focusing on metrics such as time spent on tasks, completion rates, and participation in interactive elements. Higher engagement levels often correlate with better learning outcomes.
2. Assessing academic performance by measuring improvements in academic performance by comparing grades, test scores, and overall comprehension before and after the introduction of adaptive learning technologies. This quantitative data provides a clear indicator of the effectiveness of the strategies.
3. Retention and application of knowledge through evaluating the long-term retention of learned material and the ability of students

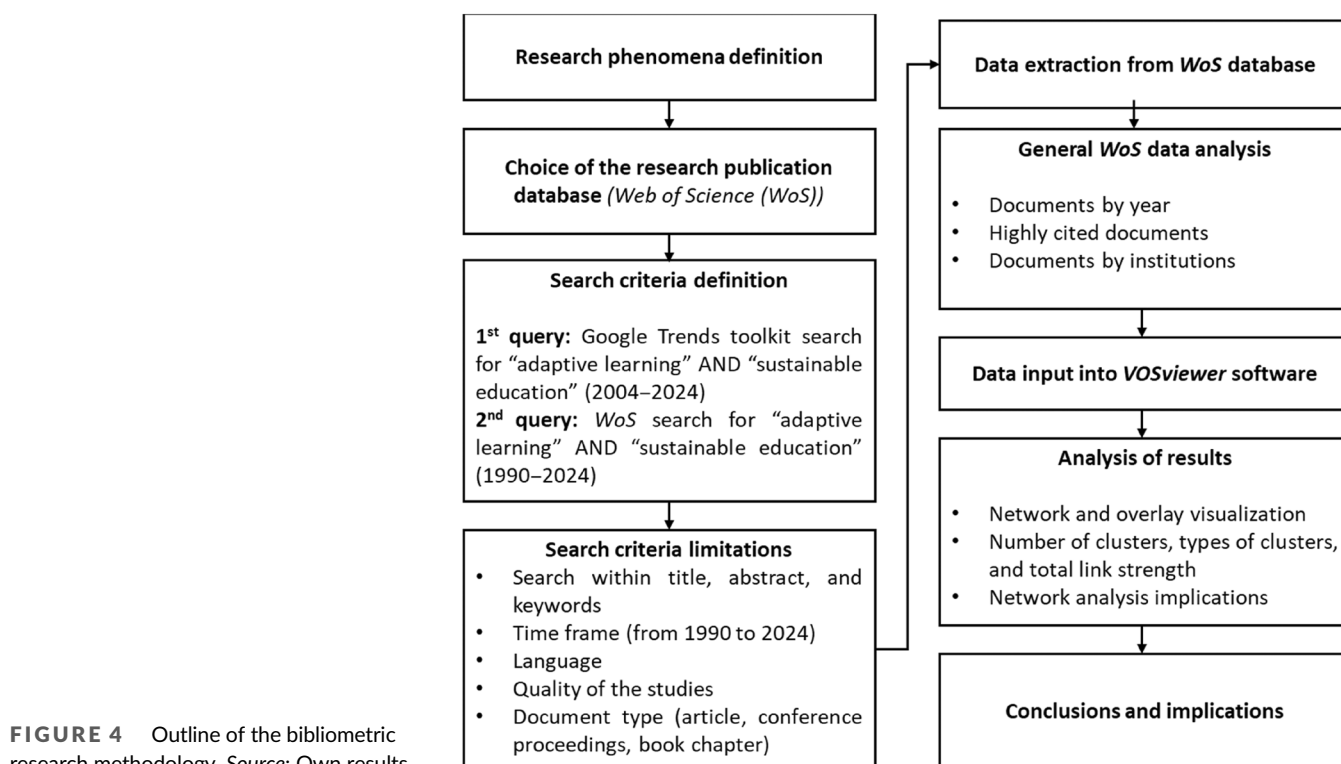


FIGURE 4 Outline of the bibliometric research methodology. Source: Own results.

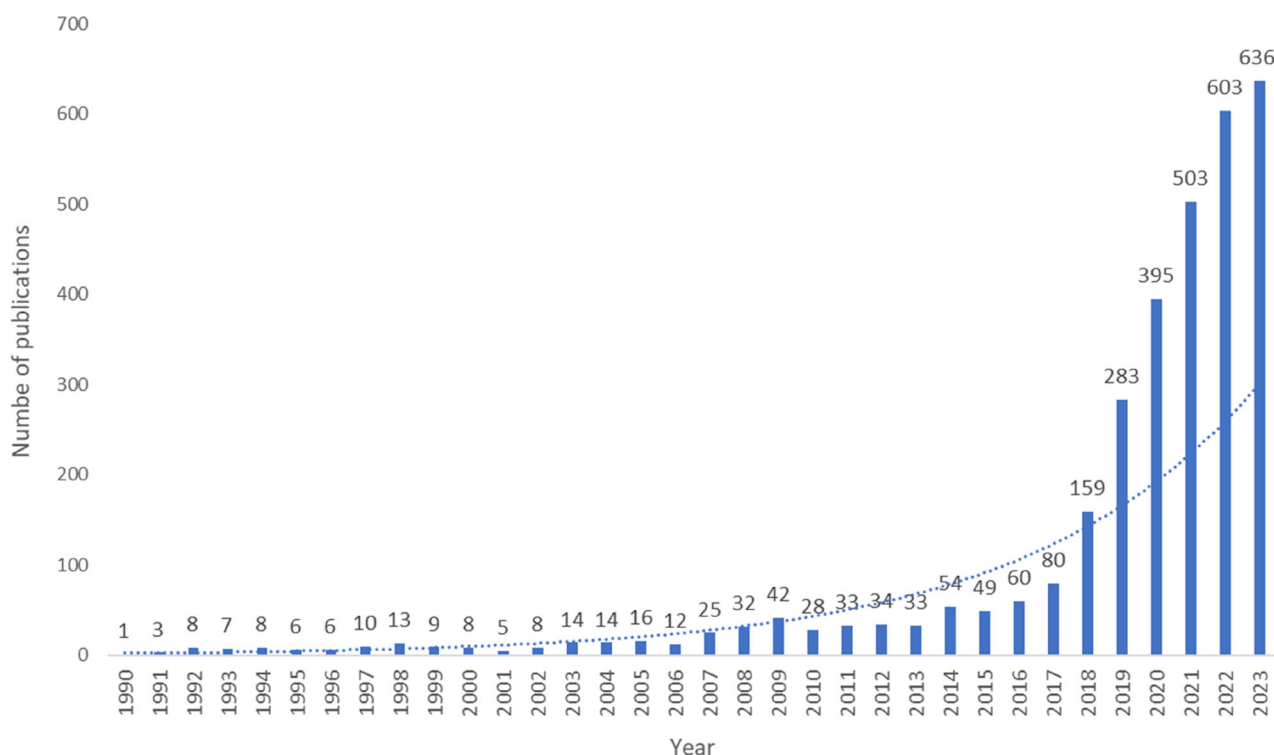


FIGURE 5 Trend of the publications on adaptive learning over time (1990–2023). *Source:* Own results.

to apply concepts in different contexts. This could involve follow-up assessments or practical applications of knowledge.

- Studying student and teacher feedback by collect qualitative data from both students and teachers through surveys and interviews to gauge their satisfaction with the adaptive learning tools and their perceived impact on the learning process.
- Increasing scalability and adaptability by examining how effectively the adaptive learning system can be scaled across various educational settings and adapted to different curricula and student demographics.
- Ensuring equity and inclusivity by ensuring that the adaptive learning technologies promote equitable educational opportunities for all students, including those from diverse socio-economic backgrounds, and do not inadvertently contribute to widening educational disparities.

By implementing this framework, educators, researchers, and policymakers can systematically evaluate the success of adaptive learning strategies, ensuring that they meet their intended goals and contribute positively to educational outcomes.

Figure 6 that follows offers an outline of the top 10 most prolific journals publishing articles and proceedings papers on adaptive learning and AI between 1990 and 2024. One can see that the share of the journals by the new publishers is rising compared to the share of the entrenched journals published by the established publishers (see Figure 6).

Furthermore, Figure 7 provides an account of the top 10 most prolific institutions producing publications on adaptive learning and AI

between 2000 and 2024. One can see very prestigious institutions such as University of London, University of California, or Chinese Academy of Sciences being among them (see Figure 7).

In addition, Table 3 provides a list of most cited publication on the topics under the investigation that has been carried out in this study (see Table 3 below).

One can observe from Table 3 that even though some of the most-cited publications are (logically) quite old and some of them are review studies, the focus that can be traced is on the e-learning and student-professor interactions in the classroom. The issues whether the teacher can be substituted by AI in the education process or what should tutors do to cope with the new challenges are also mentioned quite often.

6.2 | Text-based network analysis

Further in this sub-section, we present the results of the empirical model based on the bibliometric analysis that employs the VOSviewer software tool. The results of the analysis are presented in the form of visual network maps that allow for deducing the key patterns and occurrences.

Figure 8 below presents the visualization of the network cluster analysis with a map based on the text data from the sample of 3518 publications indexed in WoS database from 1990 until 2024. Our results of the bibliometric network analysis demonstrate that three main clusters were identified. The analysis of using keywords and phrases in the publications retrieved from WoS revealed that the key

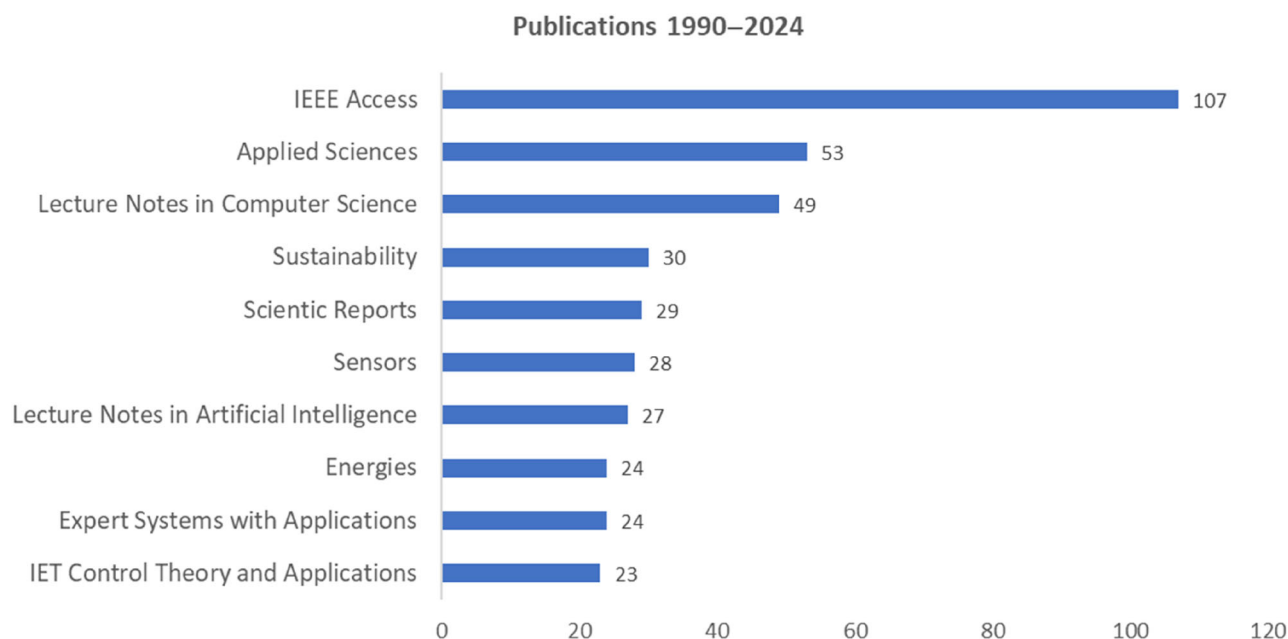


FIGURE 6 Most prolific journals (1990–2024). Source: Own results based on Web of Science (WoS).

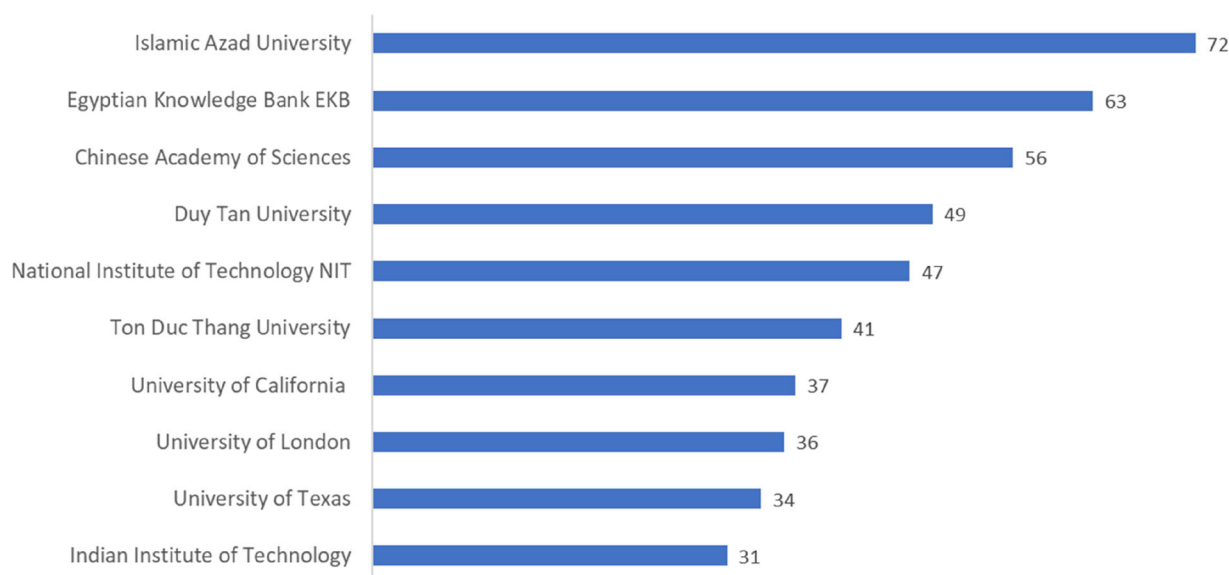


FIGURE 7 Most prolific institutions by affiliation (1990–2024). Source: Own results based on Web of Science (WoS).

terms connected to “adaptive learning” and “artificial intelligence” are most often associated with the following concepts: (i) Intelligence learning (Cluster 1 or red clustering); (ii) Parameter prediction (Cluster 2 or green clustering); and (iii) Dataset classification (Cluster 3 or blue color clustering).

Red clustering “intelligence learning” offers a panoramic view of the synergies between adaptive learning and AI. The cluster analysis reveals intricate connections between various thematic elements, showcasing their symbiotic relationship and illuminating the multifaceted dimensions underpinning the advancement of these domains. Within this cluster, keywords such as “intelligence” and “learning”

stand as linchpins, anchoring diverse sub-themes like “technology,” “learner,” “adaptation,” and “environment.”

The analysis of the “intelligence learning” cluster elucidates several significant themes. The integration of AI technology into adaptive learning settings underscores the transformative potential to create intelligent learning environments. As Grassini (Grassini, 2023) points out this convergence facilitates personalized instruction, adapting content and delivery methods to meet individual learner preferences and needs. Moreover, the analysis highlights the centrality of “interaction” and “user” elements, accentuating the importance of human-computer interaction and user-centric design in optimizing the

TABLE 3 Top 10 most cited articles (1990–2024).

Authors	Title	Number of citations
Zawacki-Richter et al. (2019)	Systematic review of research on AI applications in higher education—where are the educators?	522
Xie et al. (2019)	Trends and development in technology-enhanced adaptive/personalized learning: A systematic review of journal publications from 2007 to 2017	205
Magnisalis et al. (2011)	Adaptive and intelligent systems for collaborative learning support: a review of the field	127
Mousavinasab et al. (2021)	Intelligent tutoring systems: a systematic review of characteristics, applications, and evaluation methods	119
Almohammad et al. (2017)	A survey of artificial intelligence techniques employed for adaptive educational systems within e-learning platforms	101
Tang et al. (2023)	Trends in artificial intelligence-supported e-learning: a systematic review and co-citation network analysis (1998–2019)	88
Zhai et al. (2021)	A review of Artificial Intelligence (AI) in education from 2010 to 2020	83
Ciolacu et al. (2018)	Education 4.0-Artificial Intelligence assisted higher education: early recognition system with machine learning to support students' success	57
Ally (2019)	Competency profile of the digital and online teacher in future education	49
Tsai et al. (2020)	Precision education with statistical learning and deep learning: a case study in Taiwan	44

Source: Own results based on Web of Science database (Zhai et al., 2021; Ally, 2019; Almohammad et al., 2017; Ciolacu et al., 2018; Magnisalis et al., 2011; Mousavinasab et al., 2021; Tang et al., 2023; Tsai et al., 2020; Xie et al., 2019; Zawacki-Richter et al., 2019).

learning experience, as for example explained in the literature by Smith (Smith, 2022).

Another prominent theme steps into to the dynamic interplay between learners and their environment. The cluster underscores the role of “adaptation” as a cornerstone, where AI-driven systems dynamically adjust the learning environment in response to learner progress and performance. As Kadaruddin (Kadaruddin, 2023) explains in relation to this finding, this dynamic adaptability enhances learner engagement and promotes optimal knowledge retention. Concurrently, the notion of “challenge” also emerges as a critical factor in the literature, as AI-enabled learning environments strategically calibrate challenges to match learners' cognitive capacities, stimulating intellectual growth while preventing cognitive overload (Ahmad, Alam, et al., 2023; Huynh-The et al., 2022).

Thence, the “intelligence learning” cluster underscores the transformative potential of AI-powered adaptive learning environments. By

leveraging cutting-edge AI technologies, educational paradigms are shifting towards learner-centric models that promote tailored instruction and dynamic adaptation. The synthesis of research within this cluster underscores the need for an ecosystem that harmonizes technology, environment, and learner interactions, ushering in a new era of intelligent, adaptable, and challenging learning experiences. The implications of this synthesis extend to pedagogical design, curricular development, and the continuous evolution of AI-augmented education (Garzón, 2021; Matthew et al., 2021).

Figure 9 that follows confirms the discussion above by offering the density visualization of the network cluster analysis map. It provides the density level of the topics of adaptive learning and AI based on the quantity and depth of research (the thicker the density color, the more research was recorded and documented on this topic or concept) (see Figure 9).

Green clustering “parameter prediction” includes the ideas that elucidate the interplay between adaptive learning and AI-driven predictive models. This cluster analysis underscores the intricate bonds that connect the diverse themes and concepts interwoven by the cluster's key keywords. Among these, “prediction” and “parameter” stand as pivotal nodes, intricately linked with concepts such as “value,” “estimation,” and advanced methodologies like “artificial neural network” and “adaptive neuro fuzzy inference.”

The analysis of the “parameter prediction” cluster unfurls several prominent themes. Central to these themes is the concept of predictive modeling, where the AI is utilized to anticipate future values or conditions based on historical data and relevant parameters (Yun et al., 2022). The integration of “artificial neural networks” and “adaptive neuro fuzzy inference” emerges as a potent avenue for enhancing predictive accuracy and generalization. According to some authors, these advanced techniques demonstrate their efficiency in refining parameter estimation and forecasting, presenting promising opportunities for educational applications (Rahman et al., 2021).

Within this cluster, the concept of “indicator” assumes significance as a tool for monitoring and measuring predictive accuracy. The incorporation of indicators allows researchers and practitioners to assess the performance of predictive models and gauge their efficacy in diverse adaptive learning contexts. This underlines the importance of robust evaluation methodologies to ensure the reliability and applicability of predictive models within educational settings (Zhong et al., 2021).

The synthesis of research within the “parameter prediction” cluster underscores the transformative potential of predictive modeling in adaptive learning. By utilizing AI-powered techniques, educators can anticipate learners' future performance, needs, and progress, thereby tailoring instruction and interventions. The convergence of “parameter prediction” techniques with adaptive learning holds the promise of enhancing personalized education, optimizing learning outcomes, and fostering efficient resource allocation (Delgado et al., 2020).

The implications of this synthesis extend to pedagogical design, assessment strategies, and policy-making. The integration of predictive models calls for a holistic approach that combines sound educational theory with cutting-edge AI methodologies (Kubsch et al., 2023). Moreover, this synthesis underscores the need for

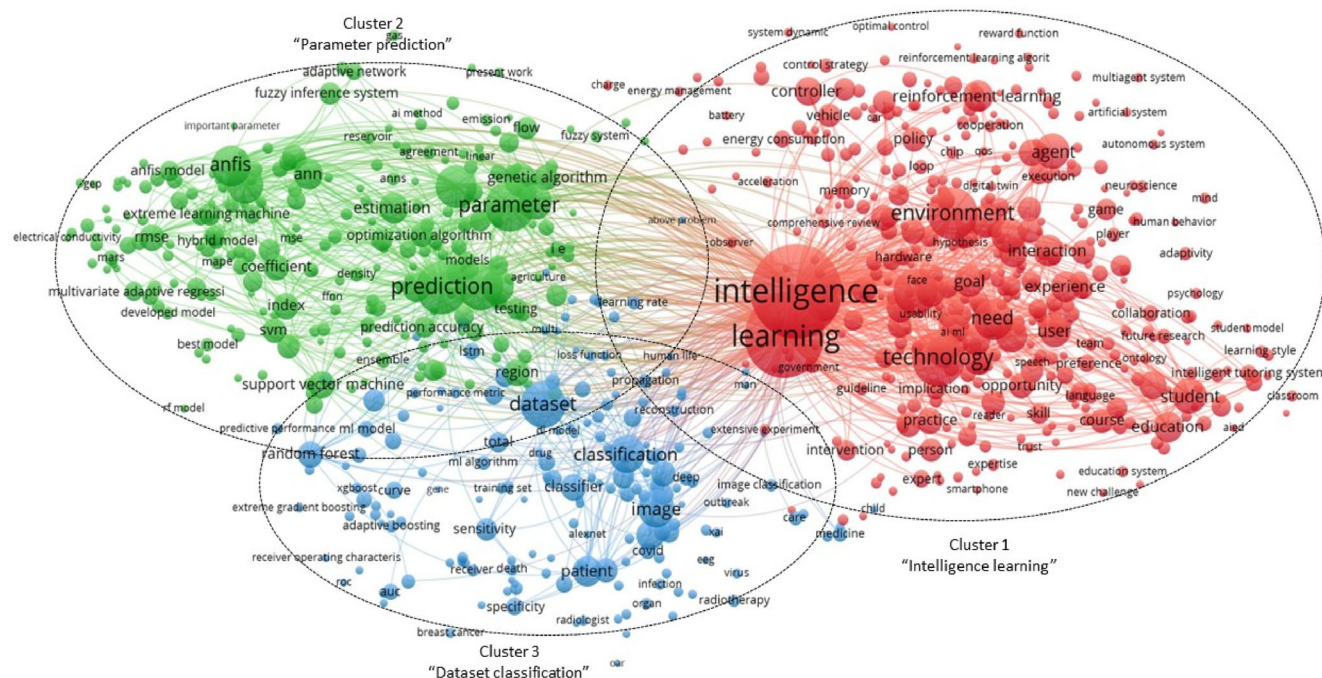


FIGURE 8 The dominant clusters of cross-sector research connected with adaptive learning and artificial intelligence retrieved from the sample of 3518 publications indexed in Web of Science (WoS). *Source:* Own results based on VOSViewer v. 1.6.18 software.

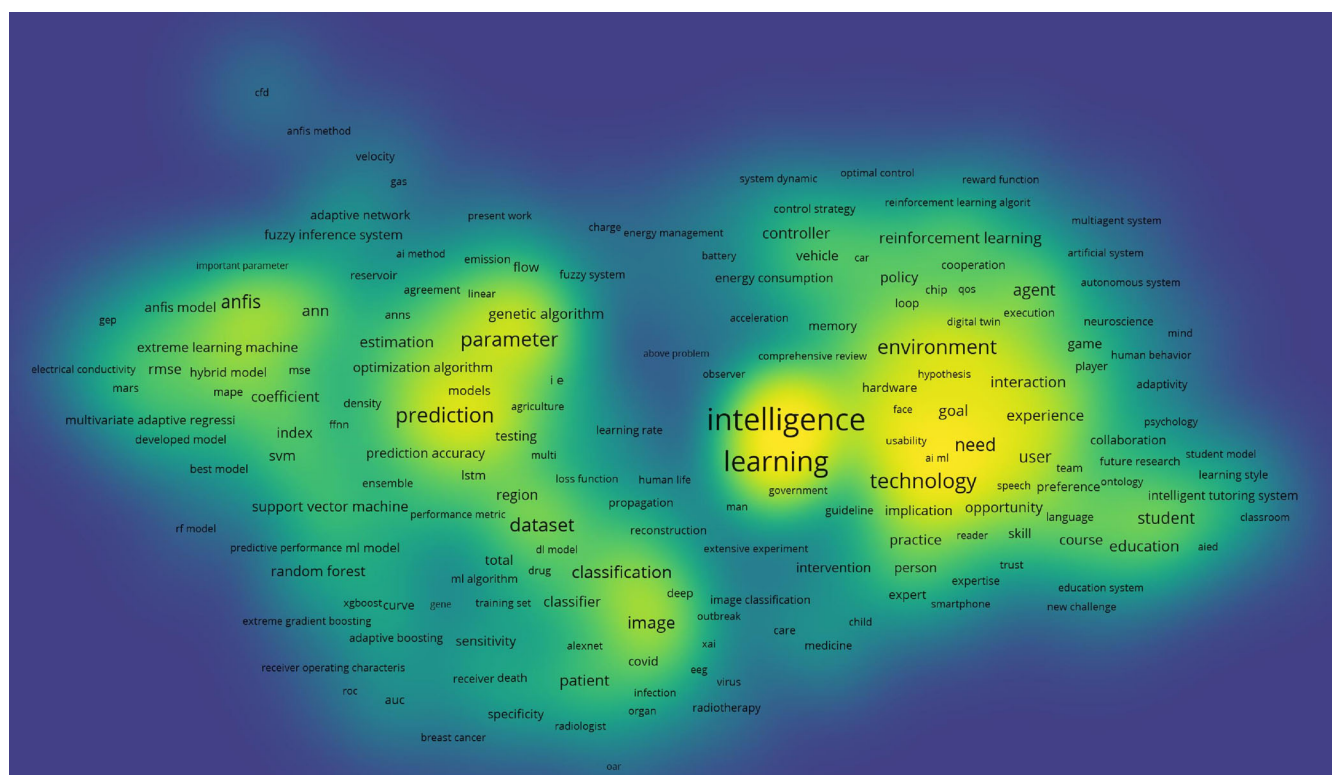


FIGURE 9 Density visualization of the network cluster analysis of the sample of 3518 publications on adaptive learning and artificial intelligence indexed in Web of Science (WoS). *Source:* Own results based on VOSViewer v. 1.6.18 software.

continued interdisciplinary collaboration, where educators, researchers, and AI practitioners converge to enhance the sophistication and applicability of parameter prediction within adaptive learning contexts (Crompton & Burke, 2023).

Finally, the blue cluster “dataset classification” encompasses a diverse range of research themes, offering a panoramic view of the interplay between adaptive learning and AI-driven dataset classification methodologies. This cluster analysis highlights the intricate

network of connections binding together various sub-themes and concepts, with focal points on “classification,” “score,” “dataset,” and methodologies such as “deep learning model” and “classifier.”

Analysis of the “dataset classification” cluster reveals several pivotal themes. Central among these is the utilization of classification methodologies to discern patterns and relationships within complex datasets. The integration of “deep learning models” and “classifiers” emerges as a potent avenue for enhancing classification accuracy, particularly in tasks related to image and data analysis. This cluster underscores the paramount importance of robust evaluation metrics, such as “score,” which enable researchers to quantify and compare the efficacy of different classification algorithms and models (Hazra et al., 2023).

The concept of “sensitivity” assumes significance within this cluster, underlining the necessity of models that can effectively capture subtle variations and nuances within datasets. The utilization of AI-driven “prediction models” within dataset classification is showcased as a powerful tool for anticipating outcomes and trends within educational contexts, allowing for data-informed decision-making and personalized instructional design (Caspari-Sadeghi, 2023).

The synthesis of research within the “dataset classification” cluster underscores the transformative potential of AI-driven classification methodologies within adaptive learning. The integration of sophisticated classification techniques empowers educators to categorize and analyze complex datasets, facilitating data-driven insights and interventions. The interplay between “dataset classification” and adaptive learning holds promise for personalized instruction, targeted interventions, and the optimization of learning experiences.

The implications of this synthesis extend to educational practice, research methodologies, and technological innovation. The amalgamation of dataset classification with adaptive learning necessitates a holistic approach that combines pedagogical insights with advanced AI techniques. Furthermore, interdisciplinary collaboration between educators, researchers, and AI practitioners is essential to ensure the refinement and application of dataset classification methodologies within adaptive learning scenarios (Minn, 2020).

6.3 | Bibliographic network analysis

Furthermore, Figure 10 reveals the results of the network map based on the bibliographic data (keyword co-occurrences, citation, and bibliographic coupling).

It is clear from Figure 10 that AI (depicted in red clustering) and machine learning (blue clustering) are mentioned in the research literature as the key methods of prediction, detection, and two-way communications for organizing e-learning (red clustering) with the help of neural networks (blue clustering).

AI and ML found an array of usages in education, especially when it comes to prediction, detection, and facilitating interactions in adaptive learning and e-learning. Additionally, the symbiotic integration of AI, ML, and neural networks fostered the evolution of predictive modeling, anomaly detection, and interactive learning interfaces

within adaptive learning and e-learning environments. These technologies redefined the boundaries of personalized education, offering tailored support, real-time interactions, and comprehensive feedback. As the educational landscape continues to evolve, the fusion of AI, ML, and neural networks promises to remain a transformative force, shaping the future of adaptive learning and e-learning paradigms. Through interdisciplinary collaboration and innovative research, this neural network-powered paradigm holds the potential to unlock new dimensions of effective and engaging educational experiences.

In summary, through the use of cluster analyses conducted in this section, it becomes evident that AI technology holds substantial importance in the context of adaptive learning, and the introduction of novel AI advancements contributes to the emergence of exceptional scholarly investigations. Through the process of keyword clustering, the foundational technological underpinnings and shared application contexts within the domain of adaptive learning can be delineated. Undoubtedly, the ongoing progression and integration of nascent technologies have significantly augmented both the scope and intricacy of potential scenarios for adaptive learning applications, progressively assuming a pivotal role in propelling advancements within this discipline.

Our results address the research questions comprehensively by demonstrating the transformative potential of AI-powered adaptive learning, acknowledging associated challenges, and indicating key trends and directions for future research and policy development in sustainable adaptive education. These results offer a forward-looking perspective, aligning the study's findings with the broader vision of personalized, ethical, and engaging education facilitated by AI and adaptive learning technologies.

7 | DISCUSSION OF RESULTS

Thus, as technology continues to evolve, further exploration and interdisciplinary collaborations hold the key to unlocking the full potential of AI-augmented adaptive learning for learners, educators, and researchers alike. When it comes to tackling RQ1, our results demonstrate that adaptive learning technologies, powered by AI, are transforming education fundamentally (Cheung et al., 2021; Ouyang & Jiao, 2021). They emphasize the personalization of education through tailored content delivery, pacing adjustments, real-time feedback, and intelligent algorithms. The integration of VR technology in classrooms, guided by AI, is highlighted as a groundbreaking advancement. These advancements underline how AI and adaptive learning are shaping the future of education, aligning with the vision of sustainable education through personalized, engaging, and experiential learning (George & Wooden, 2023; Kabudi et al., 2021; Sharma et al., 2021).

In order to fully understand the impact of adaptive learning technologies, it is crucial to incorporate feedback from students and educators who have directly experienced these systems. Student feedback can reveal how these technologies influence engagement, motivation, and learning outcomes, while educators can provide insights into the practical challenges of integrating adaptive tools into

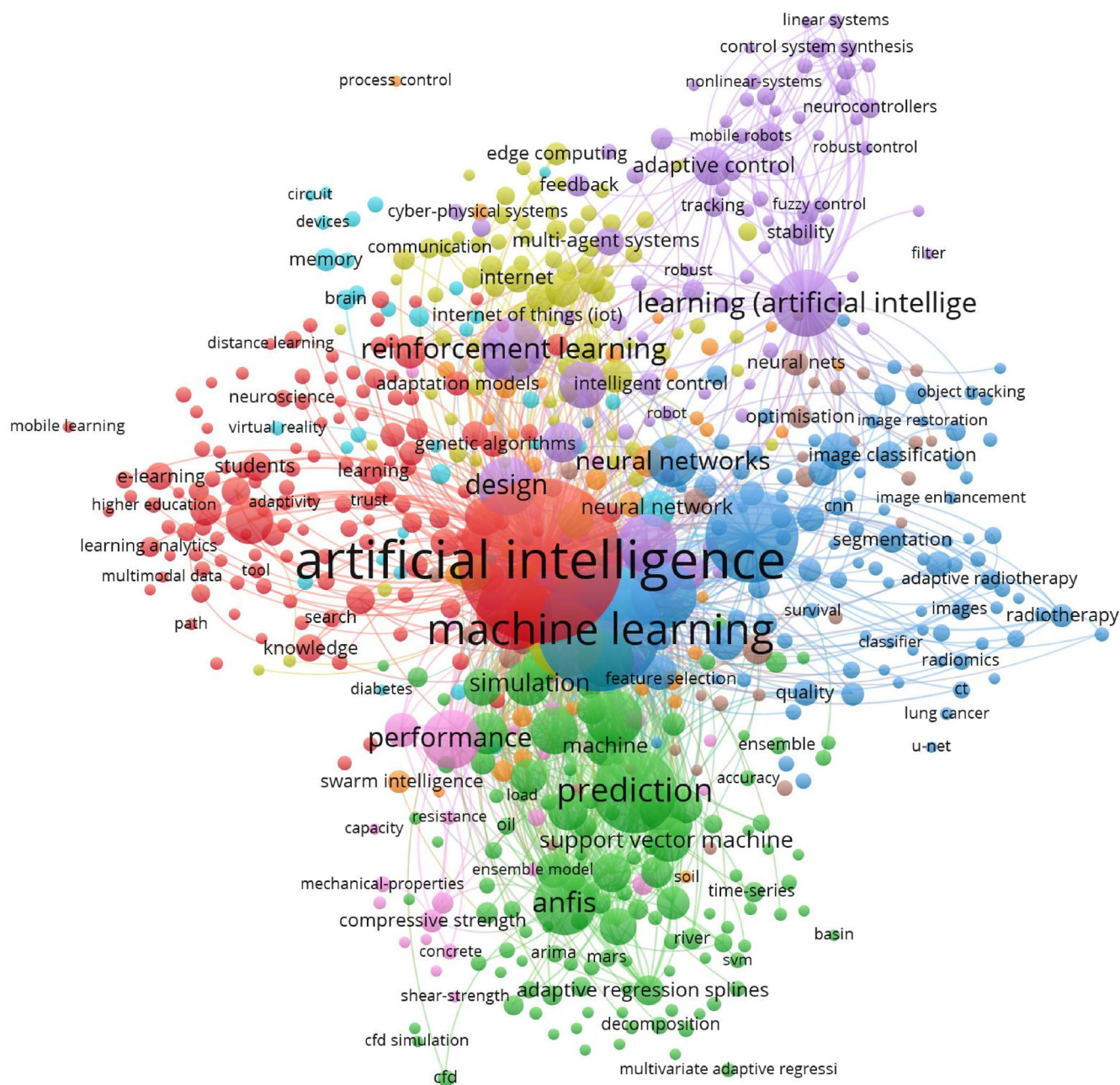


FIGURE 10 Network map based on the bibliographic data of the sample of papers containing the keywords “adaptive learning” and “artificial intelligence” retrieved from the sample of 3518 publications indexed in Web of Science (WoS). *Source:* Own results based on VOSViewer v. 1.6.18 software.

traditional classroom settings (Cavanagh et al., 2020). This real-world feedback is invaluable for validating the theoretical benefits of adaptive learning technologies and identifying areas where further refinement is needed to better meet the needs of all users. By including the voices of those who interact with these technologies daily, we can ensure that adaptive learning systems are effectively tailored to enhance educational experiences across diverse contexts.

However, it is essential to acknowledge the limitations and challenges associated with these technologies. Implementing adaptive learning technologies can be costly, requiring significant investments

in infrastructure, training, and ongoing maintenance. Data privacy and security are major concerns, as sensitive student information must be protected from breaches. Ethical considerations regarding AI decision-making in education also need to be addressed, ensuring that algorithms do not reinforce existing biases or produce unintended negative outcomes (Khreisat et al., 2024). The effectiveness of these technologies heavily depends on the quality of data and algorithms used, which can vary widely. Educators and policymakers must navigate these challenges to ensure the equitable, ethical, and effective deployment of adaptive learning technologies.

Long-term effectiveness of adaptive learning technologies on student outcomes is a critical area of investigation. Some studies demonstrated that these technologies not only improve immediate academic performance by providing personalized and adaptive learning experiences but also have a lasting impact on students' ability to retain and apply knowledge. Over time, students who engage with adaptive learning technologies often exhibit higher levels of engagement and motivation, which contributes to better educational outcomes (Yang et al., 2022). Furthermore, adaptive learning technologies help develop critical thinking and problem-solving skills by offering tailored challenges and feedback, preparing students for future educational and career success. However, to fully understand the long-term benefits, continuous research and longitudinal studies are necessary to track student progress and outcomes over extended periods.

In addition, it is also important to consider the psychological impacts of personalized learning approaches on students (Shemshack & Spector, 2020). While personalized learning can enhance motivation and engagement by allowing students to work at their own pace and focus on areas where they need the most support, it can also have potential negative effects. The constant monitoring and feedback inherent in adaptive learning systems may lead to increased pressure and anxiety for some students. Moreover, if the learning experience becomes too individualized, students may feel isolated or miss out on valuable peer interactions that contribute to their social and emotional development. To maximize the benefits of personalized learning, educators must be mindful of these psychological impacts and take steps to create a supportive and balanced learning environment.

When addressing RQ2, the challenges associated with implementing adaptive learning technologies, represent privacy concerns or ethical implications of AI algorithms. The authors recognize the need for policies and ethical standards focusing on data privacy, security, and AI deployment which is also reflected in the current research literature (Borenstein & Howard, 2021; Flores-Vivar & García-Peñalvo, 2023; Holmes et al., 2021). Our study emphasizes the importance of balancing personalized learning experiences with safeguarding sensitive student information. It also highlights the importance of equitable access, emphasizing the necessity for all students, regardless of socioeconomic background, to benefit from AI-driven educational opportunities just as other relevant studies in this field (Chan, 2023; Roshanaei et al., 2023).

Finally, the question posed in RQ3 can be answered by offering an interplay between predictive modeling, parameter estimation, and adaptive learning. Our results emphasize that as adaptive learning systems mature alongside AI technology, there is a tremendous potential to transform education, which is in accord with the outcomes of the relevant studies in this field (Alenezi, 2021; Troussas et al., 2022; Valdés & Cerdá Suárez, 2021).

All of the above significantly contributes to the understanding of the transformative role of AI and adaptive learning technologies in education. Our analysis underscores that these technologies not only personalize learning but also align with the broader goals of sustainable education, particularly in the post-COVID era (Deroncele-Acosta

et al., 2023; Whalley et al., 2021). This alignment is evidenced by the tailored content delivery and real-time feedback mechanisms, which are instrumental in creating a more dynamic and responsive educational environment (Liu & Yu, 2023). However, the challenges identified, particularly concerning data privacy and ethical AI usage, highlight the need for a comprehensive policy framework that ensures the safe and equitable implementation of these technologies. The insights gained from the bibliometric analysis underscore the evolving nature of this field and point towards predictive modeling and parameter estimation as key areas for future research (Dogan et al., 2023; Luan et al., 2020). These findings not only echo the existing literature's emphasis on the need for ethical AI but also pave the way for policy development that could shape the future landscape of sustainable adaptive education. The potential of AI in revolutionizing education, as revealed by this study, underscores the importance of addressing these challenges to fully harness the benefits of technology-driven educational models (Adiguzel et al., 2023; Liu et al., 2023).

Nevertheless, however promising all of these technological advancements in education may be, it is crucial to address concerns around privacy protection and ethical use of data in educational settings involving AI (Dhirani et al., 2023; Kooli, 2023; Marshall et al., 2022). Striking a balance between personalization and safeguarding privacy will be essential as we navigate this new era in education (Hagadone-Bedir et al., 2023; Raj & Renumol, 2022). It becomes apparent that as adaptive learning systems continue to mature alongside advancements in AI technology, the future holds tremendous potential for transforming education as we know it today.

8 | CONCLUSIONS

Overall, our results reveal the transformative potential of AI-powered parameter prediction techniques in shaping the future of education. With the further enhancement of the adaptive learning, the integration of predictive models emerges as a potent tool for tackling the power of AI to optimize and tailor educational experiences for learners around the world. The intricate dynamics between classification methodologies, dataset analysis, and adaptive learning offer a potent avenue for further implementing AI for enhancing education, optimize learning experiences, and pave the way for data-informed educational transformations led by the policy-makers and stakeholders in the field of education.

Furthermore, it becomes clear that nowadays adaptive learning represents a groundbreaking approach in personalized education that make a good use of the power of technology and AI to optimize learning experiences for individual students. By leveraging data-driven insights and intelligent algorithms, adaptive systems offer tailored content delivery, pacing adjustments, and real-time feedback that enhance both students' engagements and achievements. As this revolution in education continues to unfold, the potential for adaptive learning to transform traditional classroom practices remains immense.

As we witness the rapid rise of adaptive learning and AI in education, it becomes apparent that the future of education will be significantly shaped by these technologies. With their ability to personalize learning experiences, offer real-time feedback, and provide targeted support, AI-powered tools hold immense potential to revolutionize education in the coming years. One key aspect of the future of education lies in the creation of personalized learning paths for students. By leveraging AI algorithms and adaptive learning platforms, educators can tailor educational content and experiences to match each student's unique needs, abilities, and interests. Furthermore, AI can greatly enhance assessment methods by providing more accurate and comprehensive evaluations.

Traditional exams often fall short in measuring a student's true understanding and mastery of a subject. However, with machine learning algorithms analyzing vast amounts of data collected from various sources like quizzes, assignments, discussions, or even facial expressions during online classes, educators can gain deeper insights into students' progress and adapt teaching strategies accordingly. Another exciting prospect is the integration of VR and AR technology into classrooms. VR can create immersive environments that simulate real-world scenarios or historical events. Combined with AI-driven adaptive learning systems that adjust content based on individual student reactions or performance metrics gathered within VR simulations—students will have unprecedented opportunities for experiential learning. This is becoming a particularly interesting field for exploring and experimenting given the recent presentation of Apple Visio Pro, an AR headset created by Apple in 2023 that offers an immersive working experience but without having to be locked in the VR settings. The headset which opens up the new era of the so-called “spatial computing” (the next logical step after personal computing and mobile computing) will be on sale starting from the beginning of 2024 with a promise of unimaginable learning and presenting solutions it might deliver for sustainable education.

9 | PRACTICAL IMPLICATIONS, LIMITATIONS, AND PATHWAYS FOR FURTHER RESEARCH

When it comes to the practical implications of our research, it is quite apparent that the ascent of adaptive learning and AI technologies in education offers a paradigm shift, necessitating careful policy considerations. As these technologies take center stage, policymakers must contemplate a holistic framework that encourages the responsible integration of AI into educational practices. The development of guidelines and ethical standards for data privacy, security, and AI deployment in educational settings is imperative. Striking a balance between personalized learning experiences and safeguarding students' sensitive information is a paramount concern. Relevant government policies should also encompass the equitable distribution of AI-driven tools, ensuring that all students, regardless of socioeconomic background, have access to transformative educational opportunities.

Additionally, parental involvement also appears to be crucial in enhancing the effectiveness of adaptive learning technologies. When parents actively engage in their child's learning process, they can reinforce the personalized strategies provided by these technologies. By monitoring progress, offering encouragement, and creating a supportive learning environment at home, parents help ensure that adaptive learning tools are used effectively and consistently. Additionally, active parental involvement facilitates better communication between home and school, allowing for a more coordinated approach to addressing individual learning needs. Therefore, fostering parental involvement should be considered an integral component of implementing adaptive learning technologies, as it significantly contributes to the overall success and effectiveness of these tools.

One of the primary ethical concerns that also need to be discussed with regard to the adaptive learning technologies is the issue of privacy and data security. AI systems often require access to large amounts of personal data to function effectively, which raises questions about how this data is collected, stored, and used. Ensuring that students' privacy is protected and that data is securely managed is essential. Another ethical issue is algorithmic bias, where AI systems can inadvertently perpetuate or exacerbate existing inequalities if they are trained on biased data. It is crucial to implement safeguards and regular audits to detect and mitigate such biases. Transparency in AI decision-making processes is also important, as educators and students need to understand how AI systems arrive at their decisions. Additionally, the increasing reliance on AI in education could reduce the role of human educators, potentially impacting the development of social and emotional skills in students. To address these concerns, ethical guidelines and policies must be established to govern the use of AI in education, ensuring that these technologies are used in ways that are fair, transparent, and inclusive.

As far as the limitations of our study are concerned, several aspects need to be acknowledged. The study predominantly focuses on the broad implications of AI and adaptive learning without focusing deeply on the specific cultural, contextual, or socioeconomic nuances that could impact the adoption and effectiveness of these technologies. Moreover, the study's scope is primarily rooted in the bibliometric analysis based on the data from WoS, and its findings are contingent on the available research literature up to the knowledge cutoff date. Thence, the study might be not very efficient in capturing the recent developments or emerging trends in the field. Moreover, using publications from other databases (such as Scopus, Google Scholar, ResearchGate, or Dimensions) might also yield better and more holistic results. These databases predominantly index journals and publications that are in English and from Western countries, potentially excluding significant research published in other languages or regions. Furthermore, they may not cover all relevant conference proceedings, books, or publications in non-indexed journals, which could lead to a bias in the analysis. As a result, the findings may not fully capture the global scope of research on adaptive learning technologies, and there could be an overrepresentation of certain types of studies or research from specific regions. Acknowledging these

limitations is crucial for interpreting the results of the bibliometric analysis within the context of the broader literature.

Furthermore, one needs to realize that while bibliometric analysis provides valuable insights into research trends, influential publications, and key contributors in the field of educational technologies, it has certain limitations. Bibliometric data is often limited to indexed journals and may not capture relevant work published in non-indexed sources, conference proceedings, or emerging online platforms. Additionally, bibliometric analysis primarily focuses on quantitative measures such as citation counts, which may not fully reflect the quality, impact, or practical relevance of the research. Furthermore, this method can overlook the context and content of the studies, leading to an incomplete understanding of how adaptive learning technologies are implemented and evaluated in diverse educational settings.

As far as the pathways for further research are concerned, the creation of personalized learning pathways using AI algorithms and adaptive learning platforms presents an interesting topic for further exploration and studying. Extending beyond content delivery, researchers could plunge into the creation of AI-enhanced assessment methods that provide comprehensive evaluations of student understanding and progress. Additionally, the integration of VR technology within AI-driven adaptive learning systems presents an intriguing prospect for immersive experiential learning. Exploring the interplay between VR, AI, and adaptive learning could uncover new dimensions of educational engagement and comprehension.

Additionally, another interesting topic related to adaptive learning and AI involves the ethical use of data and privacy protection of teachers, learners, and students. Addressing concerns surrounding data privacy in AI-powered educational environments is of immense importance. Researchers in education should collaborate with policy-makers to establish ethical guidelines that balance the potential benefits of AI with students' privacy rights and to ensure the protection of data as well as be prepared for the scenarios involving cybercrime and data breaches.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

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